

How to Match a 997 to Its Outbound 850

A practical guide for tracing 997 acknowledgments to outbound purchase orders.

**Applies to**

X12 004010 / 005010

**Transactions**

850 / 997

**Task type**

Troubleshooting

**Estimated time**

10–15 minutes

Introduction

A 997 Functional Acknowledgment tells you whether your trading partner accepted the 850 purchase orders your company sent. The 997 only contains the control numbers from the outbound 850 envelope and not the PO numbers.

This guide shows how to trace a 997 back to the specific 850s and PO numbers it accepted or rejected. This process is also called acknowledgment reconciliation or correlation.

Scope

**IN SCOPE**

- 997s received for outbound 850s
- Group-level and transaction-level results
- Identifying the rejected PO and the error

**NOT COVERED**

- TA1 interchange acknowledgments
- 855 Purchase Order Acknowledgments (business acceptance)
- Map fixes and resend procedures

Before You Start

Make sure you have the following:



The raw 997 file from the trading partner, not a translator summary.



Access to your file archive, able to retrieve an 850 by partner ID and GS06, or by PO number (BEG03).



Working knowledge of X12 envelope segments (ISA, GS, ST).

Example 997

The following sample 997 is used in every step:

```
GS*FA*NORTHWINDSUP*ACMERETAIL*20261005*1502*7731*X*004010~
ST*997*0001~
AK1*PO*4521~
AK2*850*0001~
AK5*A~
AK2*850*0002~
AK3*DTM*4**8~
AK4*2*373*8*20261332~
AK5*R*5~
AK9*P*2*2*1~
SE*10*0001~
```

Steps

Follow these steps to match the 997 to the outbound 850s and identify the PO numbers.

1

Confirm the sender

Check the GS segment of the 997.

```
GS*FA*NORTHWINDSUP*ACMERETAIL*20261005*1502*7731*X*004010~
```

Element	Value	Meaning
GS01	FA	Functional acknowledgment
GS02	NORTHWINDSUP	Trading partner (Sender)
GS03	ACMERETAIL	Your organization (Receiver)



Expected result

You have confirmed that the 997 belongs to the expected trading partner.

2

Find the outbound group

Look at the AK1 segment. **AK1** contains the details of the functional group that is being acknowledged.

```
AK1*PO*4521~
```

Element	Value	Meaning
AK101	PO	Functional Group ID (GS01) of the outbound 850
AK102	4521	Functional Group Control Number (GS06) of the outbound 850

Search for the 850 file sent to this partner with GS06 = 4521.



Important

Do not match on a control number alone. Always match the trading partner and control number together.

3

Read each AK2 loop

AK2 contains the details of the transaction that is being acknowledged. **AK5** contains the acknowledgment result.

```
AK2*850*0001~
AK5*A~
```

Element	Value	Meaning
AK201	850	Transaction Set ID (ST01) of the outbound 850
AK202	0001	Transaction Control Number (ST02) of the outbound 850
AK501	A	Transaction Set acknowledgment code: A = Accepted; R = Rejected; E = Accepted with Errors

4

Read the error, if any

AK3 contains the segment with errors. **AK4** specifies the element with errors. See **Appendix A** for all error codes.

```
AK3*DTM*4**8~
AK4*2*373*8*20261332~
AK5*R*5~
```

Element	Value	Meaning
AK301	DTM	Segment ID with error
AK302	4	Fourth segment, counting ST as 1
AK304	8	Error code: 8 = Segment has data element errors
AK401	2	Data element's position in the segment (2 = DTM02)
AK402	373	Data element number (373 = Date)
AK403	8	Error code: 8 = Invalid date
AK404	20261332	Copy of the identified data element in error
AK501	R	Transaction Set acknowledgment code: R = Rejected
AK502	5	Error code: 5 = One or more segments in error

5 Check the group result

AK9 indicates the status of the functional group that is being acknowledged.

AK9*P*2*2*1~

Element	Value	Meaning
AK901	P	Functional Group acknowledgment code: A = Accepted; R = Rejected; E = Accepted with Errors; P = Partially accepted, at least one transaction set was rejected
AK902	2	Number of transaction sets specified in the functional group trailer (GE01) of the outbound 850
AK903	2	Number of transaction sets of the outbound 850 actually received by the partner
AK904	1	Number of transaction sets accepted by the partner



Note

Some partners send a 997 with no AK2 loops. In that case, AK9 provides only the functional-group acknowledgment result. Transaction-level status for individual 850s cannot be determined from the 997.

6 Get the PO numbers

Look up each ST02 from your archive to get its **BEG03**. Report the result by PO number.

Result

You can now report the 997 status by PO number.

ST02	BEG03 (PO number)	Acknowledgment Status
0001	4500012871	Accepted
0002	4500012872	Rejected: DTM02 invalid date 20261332



Tip

Never match on ST02 alone. It is unique only within its group, and almost every group starts at 0001.

Appendix A: 997 Error Codes

Use these tables to interpret the error codes in AK3, AK4, and AK5. Codes apply to X12 004010 and 005010 unless noted.

AK304 – Segment Syntax Error Code

What is wrong with the segment identified in AK3.

Code	Meaning
1	Unrecognized segment ID
2	Unexpected segment
3	Mandatory segment missing
4	Loop occurs over maximum times
5	Segment exceeds maximum use
6	Segment not in defined transaction set
7	Segment not in proper sequence
8	Segment has data element errors

AK403 – Data Element Syntax Error Code

What is wrong with the element identified in AK4.

Code	Meaning
1	Mandatory data element missing
2	Conditional required data element missing
3	Too many data elements
4	Data element too short
5	Data element too long
6	Invalid character in data element
7	Invalid code value
8	Invalid date
9	Invalid time
10	Exclusion condition violated

Later X12 versions add codes (for example, AK403 codes 12 and 13 for repeating and composite elements). Always confirm against your trading partner's implementation guide.

AK502 – Transaction Set Syntax Error Code

Why the transaction set was rejected. Appears in AK5 after the acknowledgment code.

Code	Meaning
1	Transaction set not supported
2	Transaction set trailer missing
3	Transaction set control number in header and trailer do not match
4	Number of included segments does not match actual count
5	One or more segments in error
6	Missing or invalid transaction set identifier
7	Missing or invalid transaction set control number
23	Transaction set control number not unique within the functional group

Codes 8–17 and 24–27 relate to X12 security segments and are rarely used.

Need Help?

Found an issue or have a question?

Open an issue in the [edi-integration-playbook](#) repository on GitHub.

Document History

Version	Date	Author	Change
1.0	October 2026	alowinarp	Initial release